

Assignment - 3

1) Determine the multiplication inverse modulo 26 for each of the following integers.

3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

i) 3^{-1}

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$1 = 3 + 2(-1)$$

$$1 = 3 + (26 + 3(-8))(-1)$$

$$1 = 3 + 26(-1) + 3(8)$$

$$3(8) + 26(-1) = 1$$

$$\boxed{3^{-1} = 9}$$

ii) 5^{-1}

$$26 = 5(5) + 1$$

$$26 = 5(-5) + 1$$

$$26 - 5 = 21$$

$$\boxed{5^{-1} = 21}$$

iii) 7^{-1}

$$26 = 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$26 + 7(-3) + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(-5) + 5(2) = 1$$

$$26 + 7(-5) + [26 + 7(-3)](2) = 1$$

$$26 + 7(-5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26 - 11 = 15$$

$$\boxed{7^{-1} = 15}$$

iv) 9^{-1}

$$26 = 9(2) + 8$$

$$9 = 8(1) + 1$$

$$8 = 26 + 9(-2)$$

$$9 + 8(-1) = 1$$

$$9 + [26 + 9(-2)](-1) = 1$$

$$9 + [26(-1) + 9(2)] = 1$$

$$9(3) + 26(-1) = 1$$

$$\boxed{9^{-1} = 3}$$

v) 11^{-1}

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$26 + 11(-2) + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$\boxed{11^{-1} = 19}$$

vi) 15^{-1}

$$26 = 15(1) + 11$$

$$15 = 11(1) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$26 + 15(-1) = 11$$

$$15 + 11(-1) = 4$$

$$11 + 4(-2) = 3$$

$$4 + 3(-1) = 1$$

$$15 + 11(-1) + [11 + 4(-2)](-1) = 1$$

$$15 + 11(-1) + 11(-1) + 4(2) = 1$$

$$15 + 11(-2) + 4(2) = 1$$

$$15 + [26 + 15(-1)](-2) + [15 + 11(-1)](0) = 1$$

$$15 + 26(-2) + 15(2) + 15(2) + 11(-2) = 1$$

$$15(5) + 26(-2) + [26 + 15(-1)](-2) = 1$$

$$15(5) + 26(-2) + 26(-2) + 15(2) = 1$$

$$15(7) + 26(-4) = 1$$

$$\boxed{15^{-1} = 7}$$

vii) 17^{-1}

$$26 = 17(1) + 9$$

$$17 = 9(1) + 8$$

$$9 = 8(1) + 1$$

$$26 + 17(-1) + [17 + 9(1)](-1) = 1$$

$$26 + 17(-1) + 17(-1) + 9(1) = 1$$

$$26 + 17(-1) + 17(-1) + 26 + 17(1) = 1$$

$$26(2) + 17(-3) = 1$$

$$26(-3) = 23$$

$$\boxed{17^{-1} = 23}$$

viii) 19^{-1}

$$26 = 19(1) + 7$$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$7 = 26 + 19(-1)$$

$$5 = 19 + 7(-2)$$

$$2 = 7 + 5(-1)$$

$$1 = 5 + 2(-2)$$

$$19 + 7(-2) + [7 + 5(-1)](1) = 1$$

$$19 + [26 + 19(-1)](2) + [26 + 19(-1)](-2) + 5(2) = 1$$

$$19 + 26(-2) + 19(2) + 26(-2) + 19(2) + [19 + 7(-2)](2) = 1$$

$$19(5) + 26(-4) + 19(2) + 7(-4) = 1$$

$$19(7) + 26(-4) + [26 + 19(-1)](-4) = 1$$

$$19(7) + 26(-4) + 26(-4) + 19(4) = 1$$

$$19(11) + 26(-8) = 1$$

$$\boxed{19^{-1} = 11}$$

$$i) 21^{-1}$$

$$26 = 21(1) + 5$$

$$5 = 26 + 21(-1)$$

$$21 = 5(4) + 1$$

$$1 = 21 + 5(-4)$$

$$21 + [26 + 21(-1)](-4) = 1$$

$$21 + 26(-4) + 21(4) = 1$$

$$21(5) + 26(-4) = 1$$

$$\boxed{21^{-1} = 5}$$

$$ii) 23^{-1}$$

$$26 = 23(1) + 3$$

$$23 = 3(7) + 2$$

$$3 = 2(1) + 1$$

$$3 = 26 + 23(-1)$$

$$2 = 23 + 3(-7)$$

$$26 + 23(-1) + [23 + 3(-7)](-1) = 1$$

$$26 + 23(-1) + 23(-1) + 3(7) = 1$$

$$26 + 23(-2) + 3(7) = 1$$

$$26 + 23(-2) + [26 + 23(-1)](1) = 1$$

$$26 + 23(-2) + 26(1) + 23(-7) = 1$$

$$26(8) + 23(-9) = 1$$

$$26 - 9 = 17$$

$$\boxed{23^{-1} = 17}$$

$$iii) 25^{-1}$$

$$26 = 25(1) + 1$$

$$26 + 25(-1) = 1$$

$$26 - 1 = 25$$

$$\boxed{25^{-1} = 25}$$

2) Choose any string of your choice to encrypt using the affine cipher with key values: multiplier 5 & additive 8

$$\rightarrow C = (P \times K_1 + K_2) \bmod 26$$

$$K_1 = 5, K_2 = 8$$

P.T = CAGAN

$$C = (6 \times 5 + 8) \bmod 26 \Rightarrow 38 \bmod 26 \Rightarrow 12 \rightarrow M$$

$$A = (0 \times 5 + 8) \bmod 26 \Rightarrow 8 \bmod 26 \Rightarrow 8 \rightarrow I$$

$$G = (6 \times 5 + 8) \bmod 26 \Rightarrow 38 \bmod 26 \Rightarrow 12 \rightarrow M$$

$$A = (0 \times 5 + 8) \bmod 26 \Rightarrow 8 \bmod 26 \Rightarrow 8 \rightarrow I$$

$$N = (13 \times 5 + 8) \bmod 26 \Rightarrow 73 \bmod 26 \Rightarrow 21 \rightarrow V$$

$$\boxed{C.T = MIMIV}$$

3) Encrypt "HI" using Affine cipher with $a=11$ & $b=5$

$$\rightarrow C = (P \times a + b) \bmod 26$$

$$H = (7 \times 11 + 5) \bmod 26 \rightarrow 82 \bmod 26 \rightarrow 4 \rightarrow E$$

$$I = (8 \times 11 + 5) \bmod 26 \rightarrow 93 \bmod 26 \rightarrow 15 \rightarrow P$$

$$\boxed{C.T = EP}$$

4) The cipher text "UCK" was encrypted using the affine function $ax + 2 \bmod 26$. Find the plain text.

$$\rightarrow C = (aP + 2) \bmod 26$$

$$a = 9$$

$$P = a^{-1}(C - b) \bmod 26$$

$$b = 2$$

$$\boxed{a^{-1} = 3}$$

$$U = 3(20 - 2) \bmod 26 \Rightarrow 54 \bmod 26 \rightarrow 2 \rightarrow C$$

$$C = 3(2 - 2) \bmod 26 \rightarrow 0 \bmod 26 \rightarrow 0 \rightarrow A$$

$$K = 3(11 - 2) \bmod 26 \rightarrow 45 \bmod 26 \rightarrow 19 \rightarrow T$$

$$\boxed{P.T = CAT}$$

5) Decrypt the cipher text using Affine cipher.
 'KDNXJUNDEXYZY' using the Key 3 (multiplication) & 4.

$$\rightarrow P = a^{-1}(c-b) \bmod 26$$

$$a = 3, b = 4$$

$$\boxed{3^{-1} = 9}$$

$$26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$3 + [26 + 3(-8)](-1) = 1$$

$$3 + 26(-1) + 3(8) = 1$$

$$26(-1) + 3(9) = 1$$

$$\boxed{3^{-1} = 9}$$

$$K = 9(10-4) \bmod 26 = 54 \bmod 26 \rightarrow 2 \rightarrow C$$

$$D = 9(3-4) \bmod 26 \rightarrow -9 \bmod 26 \rightarrow (26-9) \rightarrow 17 \rightarrow R$$

$$N = 9(24-4) \bmod 26 \rightarrow 180 \bmod 26 \rightarrow 24 \rightarrow Y$$

$$X = 9(23-4) \bmod 26 \rightarrow 171 \bmod 26 \rightarrow 15 \rightarrow P$$

$$J = 9(9-4) \bmod 26 \rightarrow 45 \bmod 26 \rightarrow 19 \rightarrow T$$

$$U = 9(20-4) \bmod 26 \rightarrow 144 \bmod 26 \rightarrow 14 \rightarrow O$$

$$W = 9(22-4) \bmod 26 \rightarrow 162 \bmod 26 \rightarrow 6 \rightarrow G$$

$$D = 9(3-4) \bmod 26 \rightarrow -9 \bmod 26 \rightarrow (26-9) \rightarrow 17 \rightarrow R$$

$$E = 9(4-4) \bmod 26 \rightarrow 0 \bmod 26 \rightarrow 0 \rightarrow A$$

$$X = 9(23-4) \bmod 26 \rightarrow 171 \bmod 26 \rightarrow 15 \rightarrow P$$

$$Z = 9(25-4) \bmod 26 \rightarrow 189 \bmod 26 \rightarrow 7 \rightarrow H$$

$$Y = 9(24-4) \bmod 26 \rightarrow 180 \bmod 26 \rightarrow 24 \rightarrow Y$$

$$\boxed{PT = (CRYPTOGRAPHY)}$$

6) VM $K_1 = 13$ (multiple), $K_2 = 4$ (even though it is not a valid key). Encrypt the string "input" & "alter" using the above key & find what is the inverse.

$$\rightarrow C = (P \times K_1 + K_2) \bmod 26$$

$$K_1 = 13, K_2 = 4$$

13 is not coprime of 26

So we cannot find inverse of 13 mod 26.

$\therefore$ it is not a valid key for affine cipher.

INPUT
 $I \rightarrow (8 \times 13 + 4) \bmod 26 \rightarrow 108 \bmod 26 \rightarrow 4 \rightarrow E$

$$N \rightarrow (13 \times 13 + 4) \bmod 26 \rightarrow 173 \bmod 26 \rightarrow 17 \rightarrow R$$

$$P \rightarrow (15 \times 13 + 4) \bmod 26 \rightarrow 199 \bmod 26 \rightarrow 17 \rightarrow R$$

$$U \rightarrow (20 \times 13 + 4) \bmod 26 \rightarrow 264 \bmod 26 \rightarrow 4 \rightarrow E$$

$$T \rightarrow (19 \times 13 + 4) \bmod 26 \rightarrow 251 \bmod 26 \rightarrow 17 \rightarrow R$$

ALTER
 $A \rightarrow (0 \times 13 + 4) \bmod 26 \rightarrow 4 \bmod 26 \rightarrow 4 \rightarrow E$

$$L \rightarrow (11 \times 13 + 4) \bmod 26 \rightarrow 147 \bmod 26 \rightarrow 17 \rightarrow R$$

$$T \rightarrow (19 \times 13 + 4) \bmod 26 \rightarrow 251 \bmod 26 \rightarrow 17 \rightarrow R$$

$$E \rightarrow (4 \times 13 + 4) \bmod 26 \rightarrow 56 \bmod 26 \rightarrow 4 \rightarrow E$$

$$R \rightarrow (17 \times 13 + 4) \bmod 26 \rightarrow 225 \bmod 26 \rightarrow 17 \rightarrow R$$

ERRER

Both the plain text will produce the same cipher text. The encryption is not one to one & decryption is not possible as 13 is not a valid key.

1) Find the inverse matrix for the key $\begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \text{ mod } 26$.

→ Determinant of key

$$= |3 \times 7 - 2 \times 5|$$

$$= |21 - 10|$$

$$= 11$$

$$11^{-1} \rightarrow 26 = 11(2) + 4 \quad 4 = 26 + 11(-2)$$

$$11 = 4(2) + 3$$

$$3 = 11 + 4(-2)$$

$$4 = 3(1) + 1$$

$$1 = 4 + 3(-1)$$

$$[26 + 11(-2)] + [11 + 4(-2)](-1) = 1$$

$$26 + 11(-2) + 11(-1) + 4(2) = 1$$

$$26 + 11(-3) + 4(2) = 1$$

$$26 + 11(-3) + [26 + 11(-2)](2) = 1$$

$$26 + 11(-3) + 26(2) + 11(-4) = 1$$

$$26(-3) + 11(-7) = 1$$

$$26 - 7 = 19$$

$$\boxed{11^{-1} = 19}$$

$$\text{Adjoint of matrix } (K) = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$K^{-1} = 19 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \text{ mod } 26$$

$$= \begin{bmatrix} 133 & -38 \\ -95 & 57 \end{bmatrix} \text{ mod } 26$$

$$K^{-1} = \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

8) Find the inverse of the following matrix where entries are considered modulo 26. $\begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$

$$\rightarrow K = \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix} \pmod{26}$$

$$D(K) = |11 \times 3 - 13 \times 2| = |33 - 26| = 7 \pmod{26}$$

$$7^{-1} \Rightarrow 26 \equiv 7(3) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$5 + [7 + 5(-1)](-2) = 1$$

$$26 + 7(-3) + 7(-2) + 5(2) = 1$$

$$26 + 7(5) + [26 + 7(-3)](-2) = 1$$

$$26 + 7(5) + 26(2) + 7(-6) = 1$$

$$26(3) + 7(-11) = 1$$

$$26(-11) = 15$$

$$\boxed{7^{-1} = 15}$$

$$\text{Adj}(K) = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$K^{-1} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26}$$

$$K^{-1} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

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9) Is the key valid? $\begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix}$

$$\rightarrow D(K) = |2 \times 21 - 13 \times 14| = |42 - 182| = -140$$

$$-140 \bmod 26 = 16$$

$$\det K = 16$$

A key to be valid its determinant value should be coprime with 26.

$\therefore$ It is not a valid key.

10) Encrypt using 2×2 Hill cipher for the plaintext "SOLVED" using $\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$

$$\rightarrow K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \Rightarrow D(K) = |3 \times 11 - 8 \times 7| = |33 - 56|$$

$$= -23 + 26 = 3$$

3 is coprime

SO|LV|ED

$$\rightarrow C = K \times P$$

$$SO \rightarrow P = \begin{bmatrix} 18 \\ 14 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 54 + 112 \\ 98 + 154 \end{bmatrix} = \begin{bmatrix} 166 \\ 250 \end{bmatrix} \bmod 26$$

$$166 \bmod 26 \rightarrow 10 \rightarrow K$$

$$250 \bmod 26 \rightarrow 20 \rightarrow U$$

$$LV \rightarrow P = \begin{bmatrix} 11 \\ 21 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 33 + 168 \\ 77 + 231 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26$$

$$201 \bmod 26 \rightarrow 19 \rightarrow T$$

$$308 \bmod 26 \rightarrow 22 \rightarrow W$$

$$ED \rightarrow P = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} = \begin{bmatrix} 12 + 24 \\ 28 + 33 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \bmod 26$$

$$36 \bmod 26 \rightarrow 10 \rightarrow K$$

$$61 \bmod 26 \rightarrow 9 \rightarrow J$$

Cipher text = KUTWKJ